

2(a)	any one from: To ensure no gossypol in any of the cows / to remove gossypol from cows or <i>idea of:</i> feeding on same diet standardises cows (for start of experiment) ;	1														
2(b)(i)	any three from: 1 the (mean) milk yields are, very close / similar, between groups ; 2 no statistical test or should carry out a, statistical test / <i>t</i>-test ; 3 overlap of SE (then no significant difference) ; 4 detail of which groups, are not / may be, significantly different ; <table><tr><th>not significantly different</th><th>may be significantly different</th></tr><tr><td>A and B</td><td>A and E</td></tr><tr><td>A and C</td><td>B and E</td></tr><tr><td>A and D</td><td>C and E</td></tr><tr><td>B and C</td><td>D and E</td></tr><tr><td>B and D</td><td></td></tr><tr><td>C and D</td><td></td></tr></table>	not significantly different	may be significantly different	A and B	A and E	A and C	B and E	A and D	C and E	B and C	D and E	B and D		C and D		3
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A and D	C and E															
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Question	Answer	Marks
2(b)(ii)	any two from: <i>enough evidence because:</i> 1 trend that as gossypol (in diet) increases, milk yield increases ; ora 2 paired data quote (to support) ; <i>not enough evidence because:</i> 3 <i>idea that:</i> no information that differences in milk yield are caused by (differences in) gossypol ; 4 <i>idea that:</i> no information on, nutritional content / AW, of , no gossypol diet / GM cottonseed ; 5 <i>idea that:</i> there is no cottonseed in A so cannot compare with (diets) B to E or There is only soybean in A so cannot compare with (diets) B to E ; 6 no statistical test or should carry out a, statistical test / <i>t</i>-test ;	2
2(c)(i)	<i>there is a relationship because:</i> as free (gossypol) intake increases, plasma (gossypol) concentration increases or shows a positive correlation ;	1

Question	Answer	Marks
2(c)(ii)	<i>allow description of diet instead of group B, etc.</i> any three from: 1 (Group) E has the highest risk (of toxicity) or (Group) C has the lowest risk (of toxicity) ; 2 sequence of risk, highest to lowest, is E → B → D → C ; 3 <i>ref. to higher free gossypol</i> and (–) isomer values for highest risks or <i>ref. to lower free gossypol</i> and (–) isomer values for lower risks ; 4 paired data quote from Table 2.2 to support mp 2 or 3 ; 5 free gossypol will be easily absorbed and more toxic (than bound gossypol) or Bound gossypol is less easily absorbed and less toxic / not toxic ; 6 <i>idea that:</i> uncertain how much free gossypol turns into bound gossypol when eaten ; 7 AVP ;	3